

# BLACKBODY RADIATION

## *Supports particle nature of wave*

Ideal body which behaves as a **perfect absorber**, that absorbs all electromagnetic radiation incident upon it. Such perfect absorbers are called **blackbody**.

## CONSTRUCTION AND WORKING

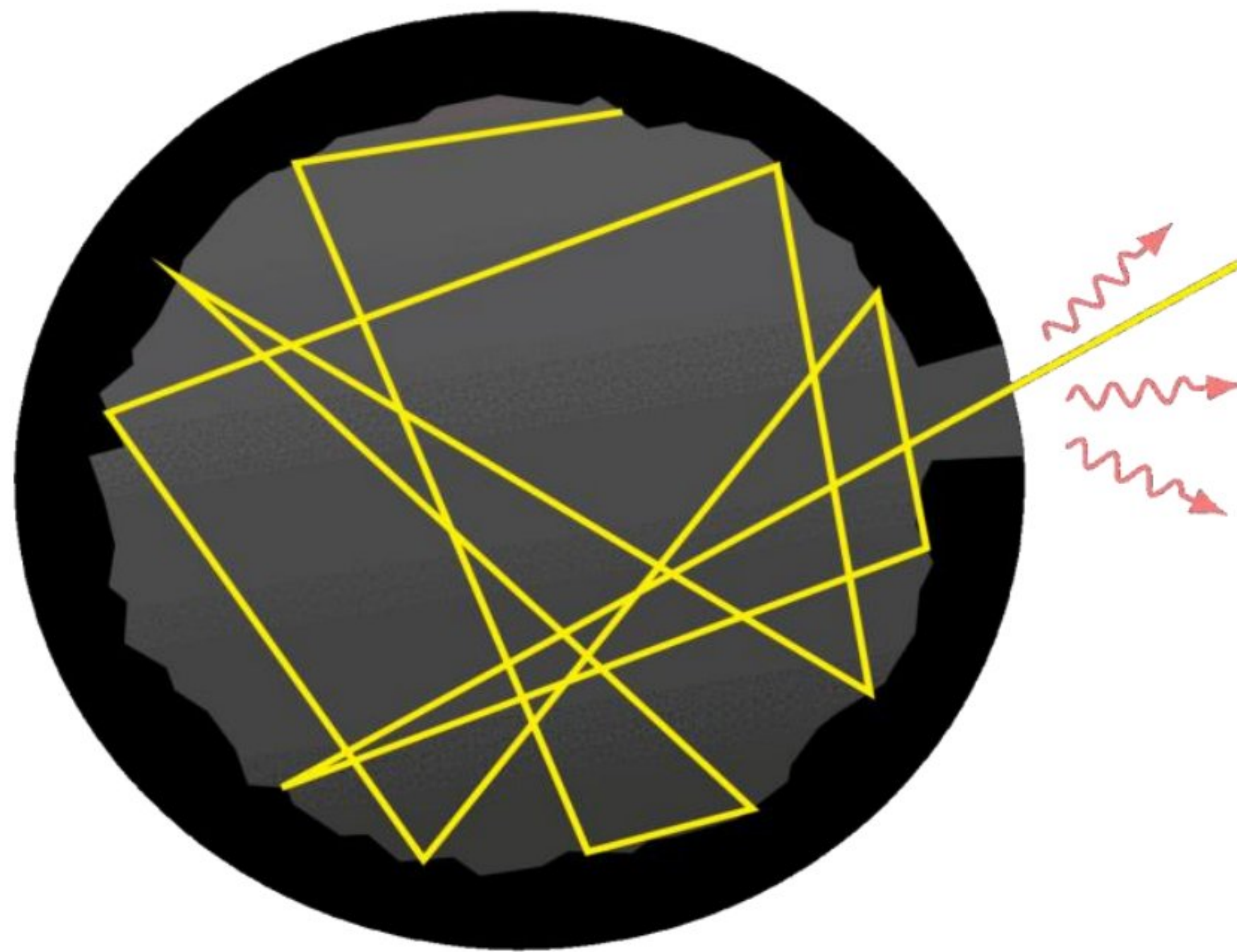
It is a hollow cavity with very small hole leading to its interior.

Any radiation entering into the cavity will be trapped inside it by **continuous reflection** from the wall which are coated with lamp black etc. and will be completely absorb.

When the cavity is heated, it radiates EMR as a result of thermal agitation (excitation) of electrons inside the cavity wall. The cavity will continuously emit and absorb the radiation and such radiations are termed as **blackbody radiation**.

Blackbody radiates more when it is hot and has its peak towards higher frequency.

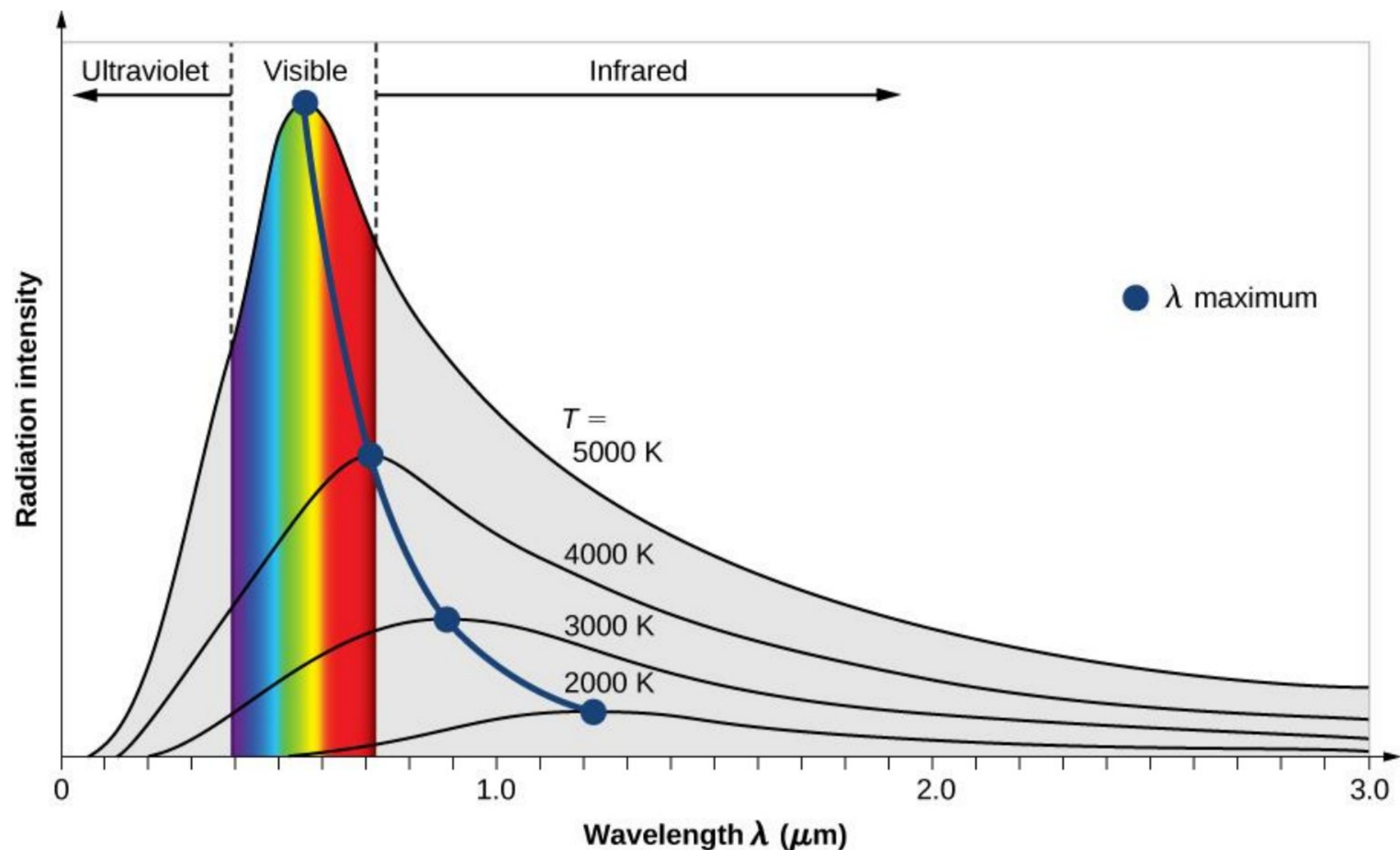




Analysis of radiation emitted can be observed by heating iron lamp, at first it will emit infrared radiation or heat waves, gradually with the increase in temperature it will glow dull red, then bright orange-red, and eventually becomes **white hot**.

The graph is plotted between energy density and wavelength, where energy density (energy stored per unit volume) is a function of frequency (oscillating electrons on the cavity walls) and temperature.





Higher the temperature, greater the amount of radiation and higher the frequency at which the maximum emission occurs. That means lowers the wavelength of emitted radiation

## CLASSICAL CONCEPT OF ENERGY DISTRIBUTION

**1.WEIN'S APPROXIMATION:** Wein's theory only holds for the distribution of energy in short wavelength (higher frequency). According to approximation, hotter the body shorter the wavelength corresponding to it.



$$\lambda_{\max} = \text{constant} \times \frac{1}{T}$$

$$\lambda_{\max} T = 0.2898 \times 10^{-2} \text{ m.K}$$

**2.RAYLEIGH JEANS THEORY:** This theory is applicable only for energy distribution in higher wavelength (lower frequency), the failure of this theory is known as ULTRAVIOLENT CATASTROPE. The theory was based on the classical concept of equipartition of energy, according to which the energy transfer between matter and radiation is continuous. At thermal equilibrium as the wavelength of the emitted light gets smaller (frequency increasing towards ultraviolet spectrum) the energy density approaches to infinity that means the body will give off radiation continuously.

### 3.PLANK'S THOERY:

According to plank's theory,

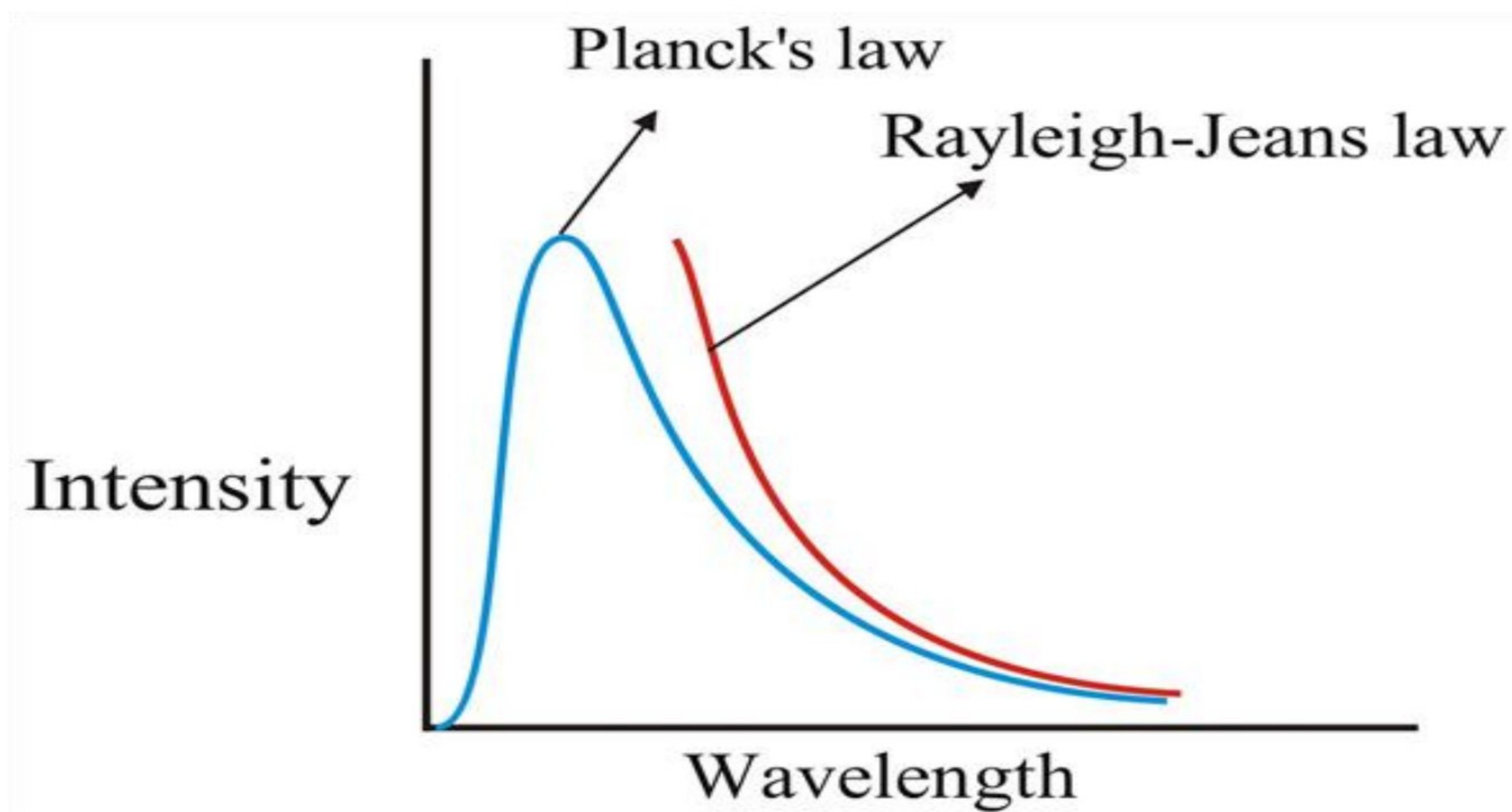
- \* Atoms at the cavity wall can emit or absorb energy in discrete values only. The smallest



amount of energy that can be absorbed or emitted in the form of electromagnetic radiation is quanta

- \* The energy of radiation absorbed or emitted is directly proportional to the frequency of radiation

$$E = nhf$$



**NUMERICAL:** What is the surface temperature of Betelgeuse, a red giant star in the constellation of Orion, which radiates the peak wavelength of about 970nm



## SOLUTION

$$T = \frac{0.2898 \times 10^{-2}}{\lambda_{\max}}$$

$$T = 2987.628 \text{ K}$$

**NUMERICAL:** The temperature of the student skin is  $33^{\circ}\text{C}$ . At what wavelength does the radiation emit from the skin reaches its peak? In which region the wavelength will be categorized?

**NUMERICAL:** A cavity radiator has its maximum spectral radiance at the wavelength of  $25\mu\text{m}$  in the infrared region of the spectrum. If the temperature of the body increases the radiant intensity of the body becomes double. Calculate the new temperature and wavelength

## SOLUTION:

Energy radiated by the body in from of EMR due to thermal vibration of molecules is given by **Stefan's Law**,

$$I = \sigma AT^4$$



$I$  = Intensity,  $T$  = Temperature,  $A$  = Surface area of the object,  $\sigma$  = Stefan-Boltzmann constant

$$\lambda_{1\max} T_1 = 0.2898 \times 10^{-2}$$

$$(25 \times 10^{-6}) T_1 = 0.2898 \times 10^{-2}$$

$$T_1 = 115.92 \text{ K}$$

Now the  $I_2$  intensity is twice of  $I_1$

$$I_2 = 2I_1$$

$$T_2^4 = 2T_1^4$$

Solving for  $T_2$

$$T_2 = 137.852 \text{ K}$$

Solving for  $\lambda_{2\max}$

$$\lambda_{2\max} T_2 = 0.2898 \times 10^{-2}$$

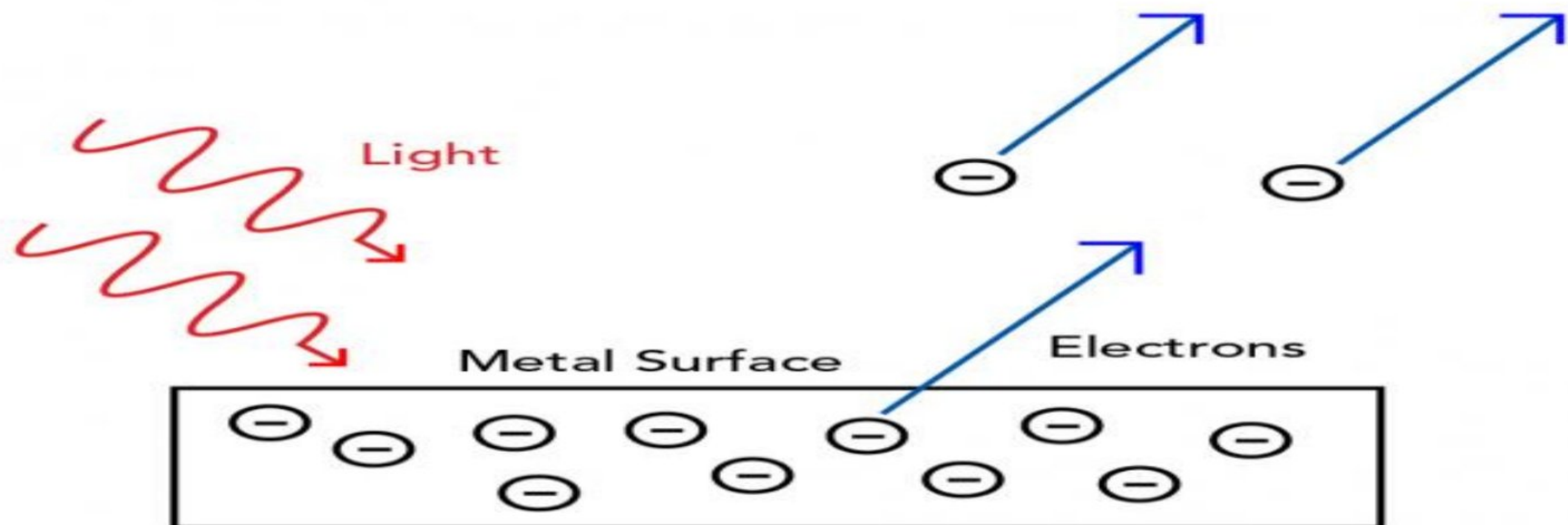
$$\lambda_{2\max} = 21 \mu\text{m}$$

**NUMERICAL:** A cavity whose wall are held at 1900K has a small hole of 1mm in diameter. At what rate does the energy escape through the hole?  
(Area= $\pi r^2$ )



# PHOTOELECTRIC EFFECT

*Supports particle nature of waves*



## Experimental results:

Einstein proposed an explanation of photoelectric effect according to utilization of photon's energy

- \* **Work-function**-The minimum amount of energy required to eject out electrons from metallic surface
- \* **KE of photon (Threshold frequency)**- Minimum amount of energy required to eject out photoelectrons, no ejection of photoelectrons below this frequency

$$\text{Photon energy} = W.F + KE_e$$

- \* **Stopping Potential**- Minimum negative potential given to the anode of the photo-



cell for which the photocurrent becomes zero

$$hf = \phi + \frac{1}{2}mv^2 \quad (1)$$

This equation is also termed as Einstein Photoelectric equation

In case if the photon energy is only sufficient to liberate the electron only, equation 1 is modified

$$hf_0 = \phi + 0$$

Substituting the value of  $\phi$  in equation 1

$$hf = hf_0 + \frac{1}{2}mv^2$$

$$hf = hf_0 + \frac{1}{2}mv^2$$

$$hf = hf_0 + eV_0 \quad (2)$$

In terms of wavelength

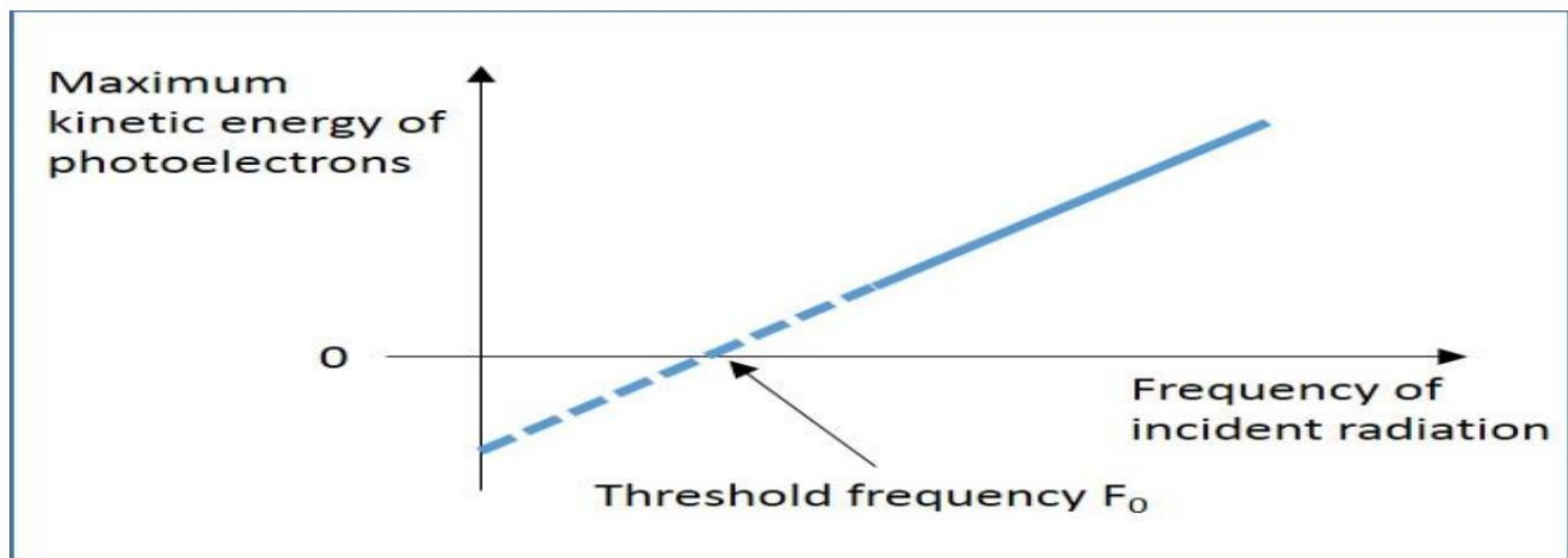
$$\frac{hc}{\lambda} = \frac{hc}{\lambda_0} + eV_0$$

Rewriting equation 2

$$V_0 = \frac{hf}{e} - \frac{\phi}{e}$$



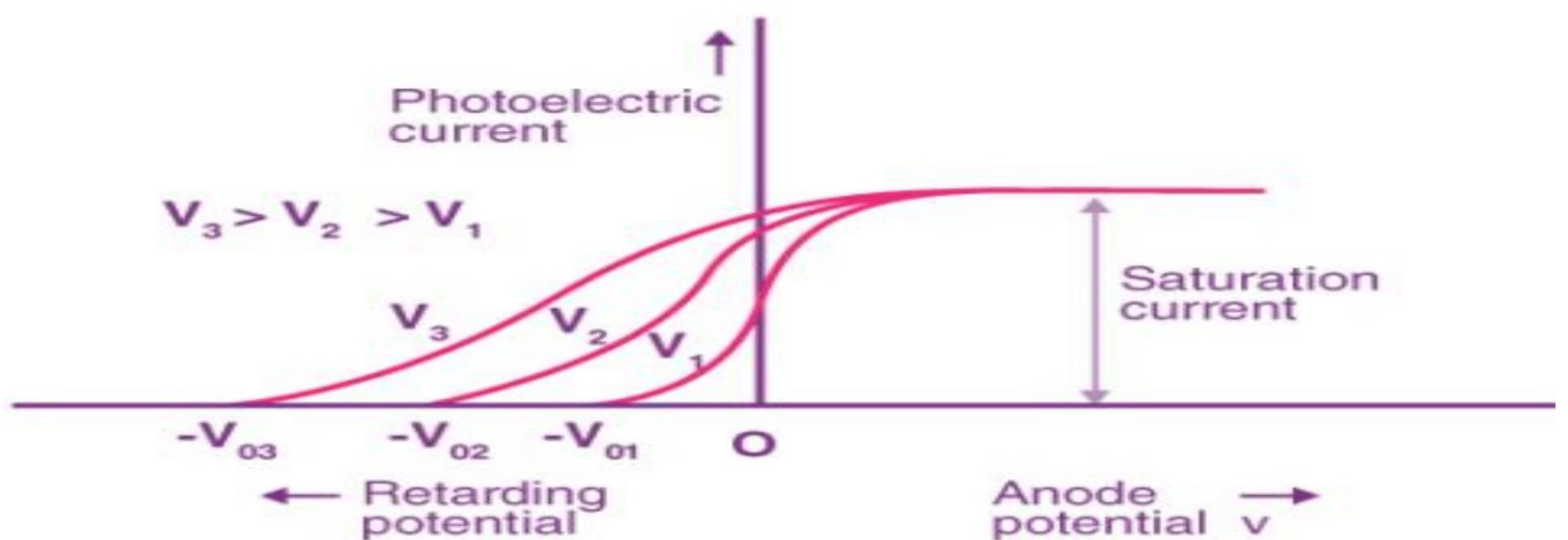
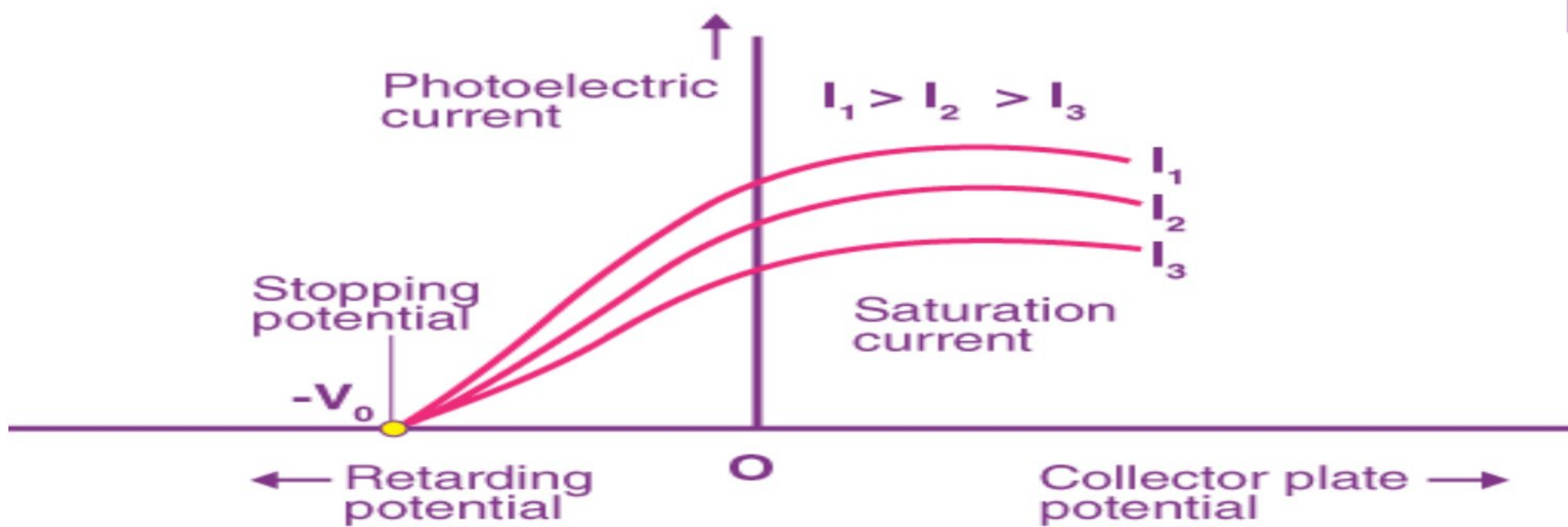
**Negative intercept** signifies that no ejection of photoelectron below threshold frequency and KE linearly increases with frequency



## Laws of photoelectric effect

1. Photo electric current is directly proportional to the intensity of incident light/no. of photons incident
2. Photons of same frequency have same energy, by increasing intensity photoelectron emission rate will increase but energy corresponding to them will remain same
3. Stopping potential is directly proportional to frequency, and independent of intensity of incident light





**NUMERICAL:** A sodium surface is illuminated with light of wavelength  $0.300\mu\text{m}$ . The work-function of the sodium is  $2.46\text{eV}$ , Calculate

- The energy of each photon in electron volts
- The minimum KE of the ejected photoelectron
- Cutoff wavelength for sodium



## SOLUTION:

a.  $E = hf$  (1)

Since  $f$  is unknown it can be easily calculated by

$$f = \frac{c}{\lambda} = \frac{3 \times 10^8}{0.3 \times 10^{-6}}$$
$$f = 1 \times 10^{15} \text{ Hz}$$

Substituting in equation 1

$$E = (6.63 \times 10^{-34})(1 \times 10^{15})$$

$$E = 6.63 \times 10^{-19} \text{ Joules}$$

$$E = 6.63 \times 10^{-19} \text{ Joules} \left( \frac{1}{1.6 \times 10^{-19}} \right)$$

$$E = 4.14 \text{ eV}$$

b.  $K.E = hf - \phi$

$$K.E = E - \phi$$

$$K.E = 4.14 - 2.46$$

$$K.E = 1.68 \text{ eV}$$

c.  $\lambda = \frac{hc}{\phi}$  (2)

$$\phi = (2.46 \text{ eV})(1.6 \times 10^{-19})$$

$$\phi = 3.94 \times 10^{-19} \text{ joules}$$

Substituting the values in equation 2

$$\lambda = 5.05 \times 10^{-7} \text{ m}$$



**NUMERICAL:** Find the work function of sodium if the cutoff frequency  $f_0 = 4.39 \times 10^{14} \text{ Hz}$  ( $\Phi_0 = hf_0$ )

(Answer: 1.82eV)

**NUMERICAL:** When two ultraviolet beams of wavelength  $\lambda_1 = 80 \text{ nm}$  and  $\lambda_2 = 110 \text{ nm}$  fall on a lead the produce photoelectrons with maximum energies 11.390eV and 7.154eV. Estimate the numerical value of the Planck constant

**SOLUTION:**

a.  $\lambda_1 = 80 \times 10^{-9} \text{ m}$ ,  $\lambda_2 = 110 \times 10^{-9} \text{ m}$

$$f_1 = \frac{c}{\lambda_1}, \quad f_2 = \frac{c}{\lambda_2}$$

$$f_1 = 3.75 \times 10^{15} \text{ Hz}, f_2 = 2.73 \times 10^{15} \text{ Hz}$$

$$E_{\text{max}1} = hf_1 - \Phi$$

$$E_{\text{max}2} = hf_2 - \Phi$$

Substituting the values and solving the equation simultaneously for h

$$11.39 \text{ eV} = h(3.75 \times 10^{15}) - \Phi \quad (1)$$

$$7.154 \text{ eV} = h(2.73 \times 10^{15}) - \Phi \quad (2)$$

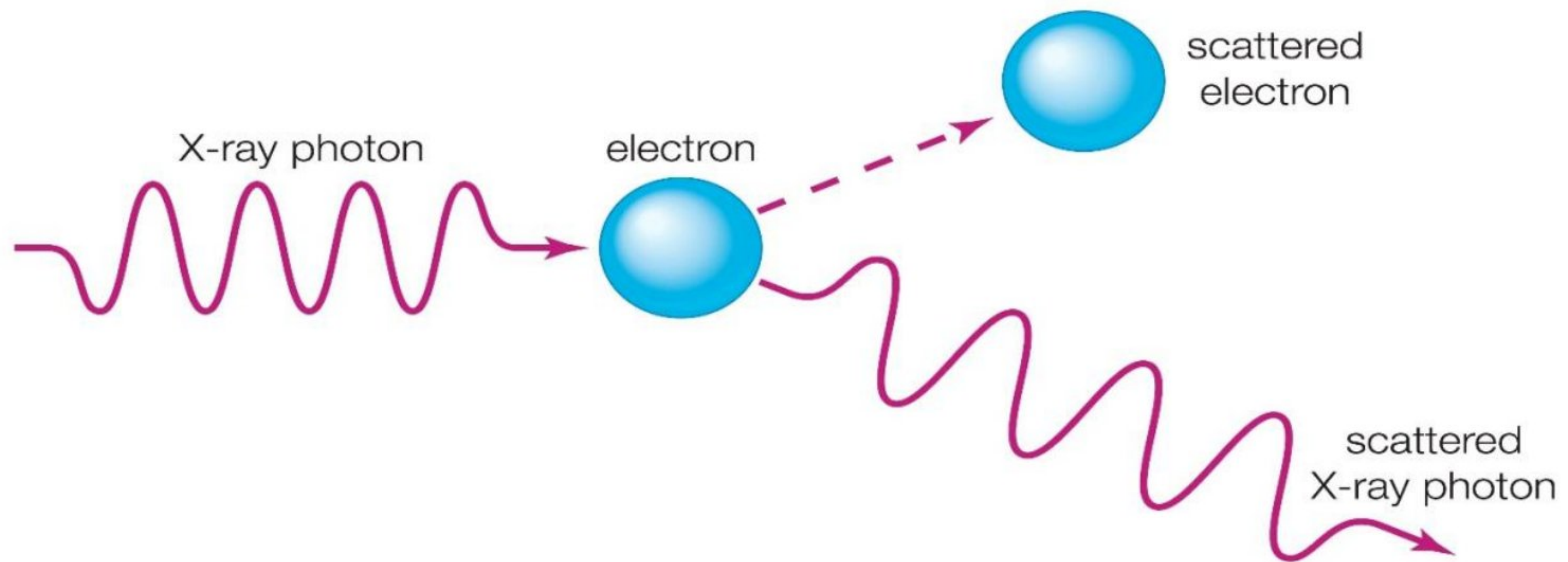
$$4.236 \times 1.6 \times 10^{-19} = h(1.02 \times 10^{15})$$

$$h = 6.64 \times 10^{-34} \text{ Js}$$



# COMPTON EFFECT

***Confirms the particle nature of wave*** ■



Inelastic scattering of photon where incoming x-ray beam behaved not as a wave but as an assembly of photons with energy ( $E=hf$ )

Scattered radiation is less energetic than incident radiation and the difference in energy is going into kinetic energy of electron

**Loss in photon energy=Gain in electron energy**

Classically, it was thought that the wavelength before and after scattering remains same. But in actual the scattered photon wavelength increases by amount  $\Delta\lambda$  (shifted wavelength)



The amount of energy reduction depends on the angle at which x-rays are scattered.

The shift in wavelength of scattered photon is

$$\Delta\lambda = \lambda - \lambda_0 = \frac{h}{m_e c} (1 - \cos\theta)$$

$\frac{h}{m_e c} = \lambda_c$  is Compton wavelength

Compton wavelength is very small as compare to visible range wavelength, so shift in wavelength will be difficult to detect if visible light is used. This only applicable for **high energy incident photons**

**NUMERICAL:** X-rays of wavelength  $\lambda_i = 0.2$  nm are scattered from a block of material. The scattered x-rays are observed at an angle of  $45^\circ$  to the incident beam. Calculate the wavelength of x-ray scattered from this angle and compute the fractional change in the energy of a photon in the collision

**SOLUTION**

**Scattered wavelength;**



$$\Delta\lambda = \frac{h}{m_e c} (1 - \cos\theta)$$

$$\Delta\lambda = \frac{(6.63 \times 10^{-34})}{(9.11 \times 10^{-31})(3 \times 10^8)} (1 - \cos 45)$$

$$\Delta\lambda = 0.000711 \text{ nm}$$

$$\Delta\lambda = \lambda f - \lambda i$$

$$\lambda f = \Delta\lambda + \lambda i$$

$$\lambda f = 0.200711 \text{ nm}$$

### Fractional change in energy

$$\frac{\Delta E}{E} = \frac{E_f - E_i}{E_i}$$

$$\frac{\Delta E}{E} = \frac{\frac{hc}{\lambda f} - \frac{hc}{\lambda i}}{\frac{hc}{\lambda i}} = \frac{\frac{1}{\lambda f} - \frac{1}{\lambda i}}{\frac{1}{\lambda i}}$$

$$\frac{\Delta E}{E} = \frac{\lambda i}{\lambda f} - 1 = \frac{\lambda i - \lambda f}{\lambda f} = -\frac{\Delta\lambda}{\lambda f}$$

$$\frac{\Delta E}{E} = -3.54 \times 10^{-3}$$



Negative sign signifies that the final energy is smaller than the initial energy of photon

**NUMERICAL:** X-ray of wavelength 10 pm are scattered from the target. Find the wavelength of x-rays scattered through  $45^\circ$ , maximum wavelength present in the scattered x-ray, maximum KE of recoil electron

### **SOLUTION**

#### **Wavelength through $45^\circ$**

$$\lambda_f = \lambda_i + \lambda_c(1 - \cos\theta)$$

$$\lambda_f = 10.711 \text{ pm}$$

#### **Maximum wavelength**

Maximum wavelength occurs when  $(1 - \cos\theta) = 2$

$$\lambda_f = \lambda_i + 2\lambda_c$$

$$\lambda_f = 15.571 \text{ pm}$$

#### **Maximum recoil energy**

$$K.E = hc\left(\frac{1}{\lambda_i} - \frac{1}{\lambda_f}\right)$$

$$K.E = 5.795 \times 10^{-15} \text{ J}$$



**NUMERICAL:** An incident 71-pm X-ray is incident on a calcite target. Find the wavelength of the X-ray scattered at a  $30^\circ$ . What is the largest shift that can be expected in this experiment?

**SOLUTION:**

The shift at  $\theta=30^\circ$

$$\Delta\lambda = \lambda_c(1 - \cos 30) = 0.314\lambda_c = 0.325\text{pm}$$

This gives the scattered wavelength:

$$\lambda' = \lambda + \Delta\lambda = (71 + 0.325)\text{pm} = 71.325\text{pm}.$$

The largest shift:

$$(\Delta\lambda)_{\text{max}} = \lambda_c(1 - \cos 180) = 2(2.43\text{pm}) = 4.86\text{pm}.$$

## DE-BROGLIE'S HYPOTHESIS

*All matter particle shows dual behavior*

**De-Broglie hypothesis:**

*Since, photons have both wave and particle characteristics, perhaps all form of matter have both properties*



All material particle with momentum 'p' should have characteristic wavelength 'λ' as De-Broglie wavelength

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

$$f = \frac{E}{h}$$

Momentum can also be defined as,

$$p = \sqrt{2mK.E}$$

Thus wavelength can be written as,

$$\lambda = \frac{h}{\sqrt{2mK.E}}$$

**NUMERICAL:** If the wavelength of the electron is  $5 \times 10^{-7} m$ , how fast the electron is moving?

**SOLUTION:**

$$V = \frac{h}{m\lambda} = 1.454 \times 10^3$$

$$V = 1.454 \text{ Km/s}$$



**NUMERICAL:** An electron and photon both has a kinetic energy of 3eV, what are their wavelength. Also comment the relation between momentum and wavelength by considering De-Broglie hypothesis.

**SOLUTION:**

Working for electron;

$$E_e = \frac{m_e v^2}{2}$$

Solving for V

$$v = \sqrt{\frac{2E_e}{m_e}} = \sqrt{\frac{(2)(3)(1.6 \times 10^{-19})}{(9.11 \times 10^{-31})}}$$

$$v = 1.026 \times 10^6 m/s$$

$$\lambda = \frac{h}{mv} = 0.708 \text{ nm}$$

Working for photon;

$$E_p = \frac{m_p v^2}{2}$$



Solving for  $v$ ,

$$v = \sqrt{\frac{2E_p}{m_p}}$$

$$v = 23957.236 \text{ m/s}$$

$$\lambda = \frac{h}{mv} = 16.535 \text{ pm}$$

**NUMERICAL:** Calculate the De-Broglie wavelength of

a. Dust particle of mass  $1.0 \times 10^{-9} \text{ kg}$  drifting with speed of  $2 \text{ cm/s}$

(Answer:  $3.3 \times 10^{-23} \text{ m}$ )

b. An electron whose KE is  $120 \text{ eV}$

(Answer:  $1.1 \times 10^{-10} \text{ m}$ )



